VoiceOps 2026 Capstone Report

ZentroMart customer support agent | Audit snapshot: 25 October 2026 | Dataset reference date: 20 October 2026

Project overview

A four-step customer support agent: Gemini prompting, document-grounded RAG with source IDs, multi-turn conversations with verified order/ticket tools, and Gnani-only speech input/output. The PowerShell CLI and local/deployed website share the same Python agent; all capstone customer records are synthetic.

STEP 1	STEP 2	STEP 3	STEP 4
LLM prompt	RAG + sources	Conversation + tools	Gnani STT / TTS

Saved full-pack evaluation

Run	Scope	Required facts	Evidence / tool checks
Step 1 prompt only	52 questions	8 / 49 (16.3%)	No RAG; tool-call screening 52 / 52
Step 2 grounded RAG	52 questions	42 / 49 (85.7%)	Evidence 47 / 47; tool calls 52 / 52
Step 3 single questions	52 questions	40 / 49 (81.6%)	Evidence 47 / 47; tool calls 52 / 52
Step 3 ASR-text simulation	52 questions	35 / 49 (71.4%)	Evidence 44 / 47; tool calls 52 / 52
Step 3 conversations, post-fix	96 turns	59 / 77 (76.6%)	Evidence 96 / 96; tool calls 96 / 96

Interpretation: saved full-pack Step 1/2 used the same `question` field; it is not a fresh full `asr_text` run. Scores are literal fact/source checks and need human review for paraphrases, dates, numbers and language quality.

Quota-limited live asr_text pilot

Run	Required facts	Gold evidence	Other checks
Step 1 baseline	1 / 3	0 / 2	Prompt-only
Step 2 before fix	1 / 3	1 / 2	Initial sample
Step 2 after fix	3 / 3	2 / 2	Hard negatives 3 / 3; tool calls 3 / 3

Cases: PQ0002 spoken-SKU warranty, PQ0003 wallet refund timing, and PQ0004 unsupported midnight delivery. This is a three-case smoke sample, not the full 52-question benchmark. Gemini calls were spaced by 4.1 seconds.

Verified fixes and remaining limits

Targeted changes from the quota-limited live pilot

Observed issue	Targeted change	Verified result
Speech-style SKU did not map to a catalog row	Normalize spoken digit forms against actual product SKUs during retrieval.	PQ0002 retrieved ELE-0012 and WAR-2; required facts and evidence passed.
Midnight delivery question invited an unsupported guess	Clarify that delivery lead-time facts do not establish slot availability.	PQ0004 abstained instead of inventing hours.
Gnani streaming WAV sentinel looked like a very long clip	Read actual WAV data length when RIFF uses the streaming-size sentinel.	Synthetic WAV batch completed; measured reply audio: 13.44 seconds. Offline unit test passes.

Voice and offline verification

Check	Result
Live voice smoke	One synthetic prerecorded WAV completed Gnani STT -> Gemini agent -> Gnani TTS (timbre-v2.5). Transcript, reply, 11 source IDs/chunks and 13.44-second reply WAV are saved under sample_outputs/live_voice_smoke_verified/.
Personal browser microphone	Not verified in this run. An earlier browser attempt returned no intelligible speech. A clear personal recording still needs to be spoken into the browser mic before presenting; no personal recording is included in the ZIP.
Offline tests	38 tests: 37 passed; one socket-dependent test skipped because local sockets are blocked in this environment.
Line coverage	81.1% (1,422 / 1,754 executable lines) across 10 first-party Python modules exercised by tests, measured using standard-library trace. Branch coverage was not measured.
Dataset validator	Passed: all 9 CSVs, 21 knowledge documents, joins, order math, practice evidence and verification gates.

Reproduce locally

Purpose	Command
Dataset validation	python validate_dataset.py
Quota-limited live evaluation (uses Gemini)	python evaluate.py --dataset dataset --steps 1 2 --limit 3 --output reports/asr-pilot.jsonl
One-file voice sample (uses Gnani and Gemini)	python batch_run.py sample_outputs/live_voice_smoke_verified/input --output reports/voice-smoke

Submission: GitHub repository | deployed website | PowerShell: `python run.py --step 3`. Keep .env, API keys, virtual environments and the supplied dataset copy out of the ZIP. Full Step 1/2 asr_text evaluation remains pending until quota permits; the personal mic check needs the user to record a clear short phrase.